

MATH FOR AI • EPISODE 0.4

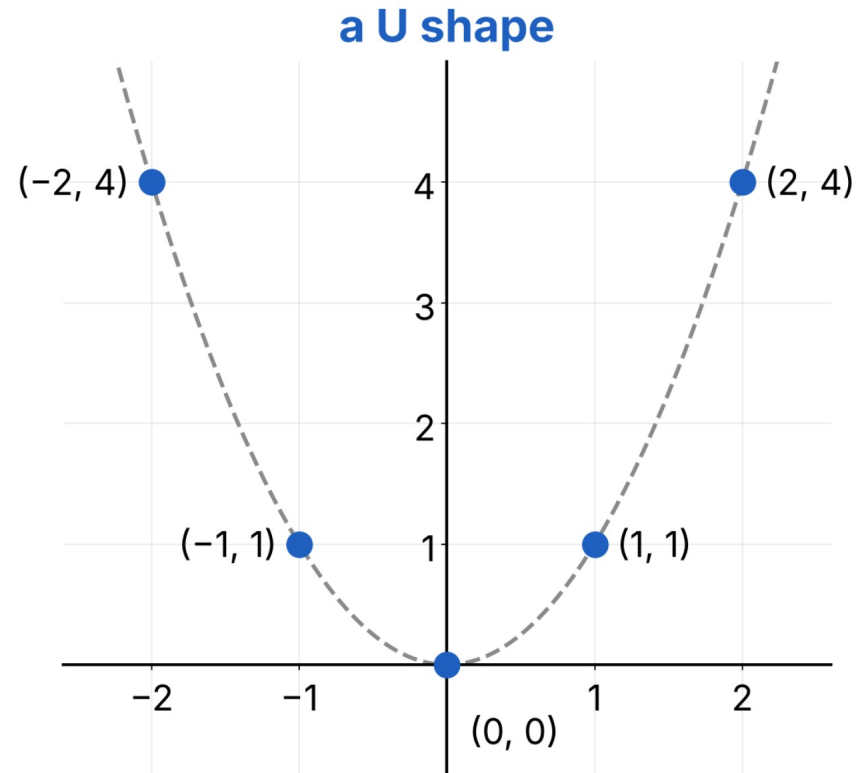
Graphing Functions

Every function draws a picture.

Dr. Joe

Last time's challenge

x	$g(x)$
-2	4
-1	1
0	0
1	1
2	4



$$g(x) = x^2$$

The outputs go 4, 1, 0, 1, 4.

Down, then back up. **A U shape.**

BONUS

$$g(f(1)) = g(3) = 9$$

You've already seen a graph



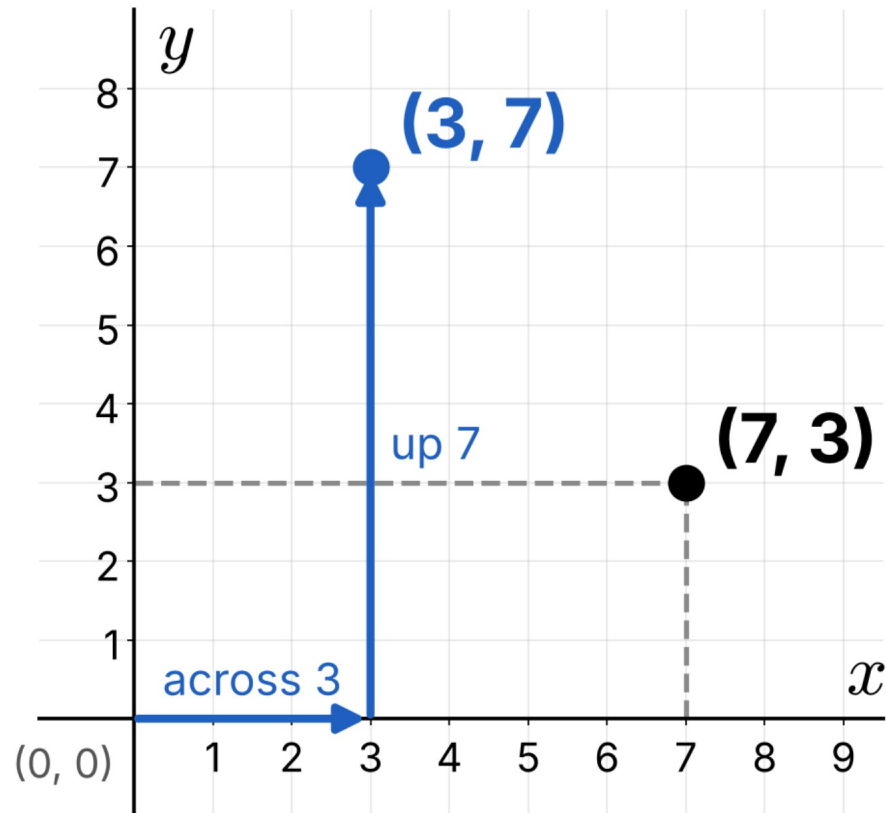
EPISODE 0.1'S ERROR VALLEY

Knob setting w goes in. Error score comes out.

Training rolled downhill on this picture.

Today: learn to read the picture.

The grid: across, then up



(x, y)

First number: how far **across**.

Second number: how far **up**.

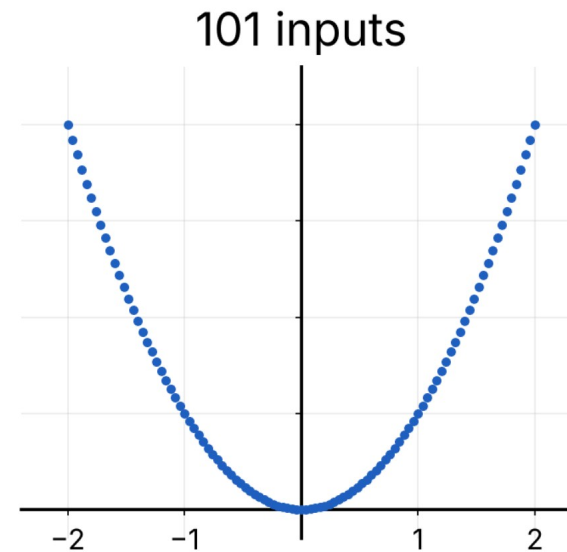
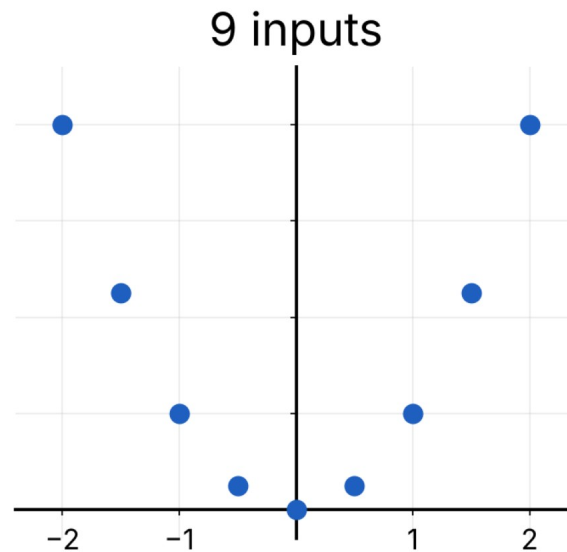
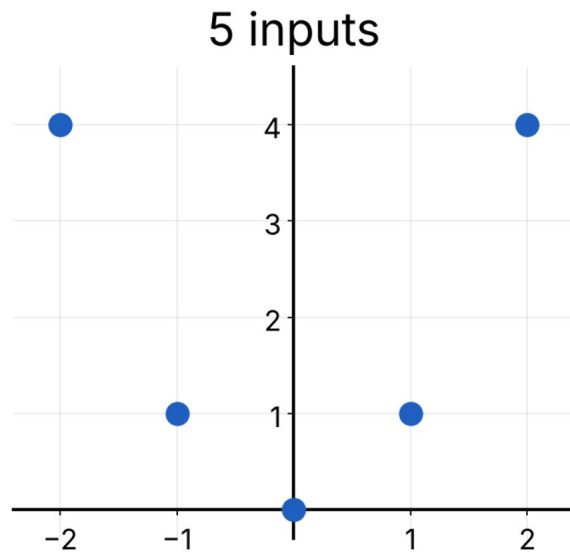
The **x-axis** runs across. The **y-axis** runs up. They cross at $(0, 0)$, the **origin**.

Order matters: $(3, 7)$ and $(7, 3)$ are different spots.

A graph is a pile of dots

Name the output y : $y = f(x)$ Every dot sits at $(x, f(x))$

$y = x^2$: plug in more and more inputs



Graphing in Python

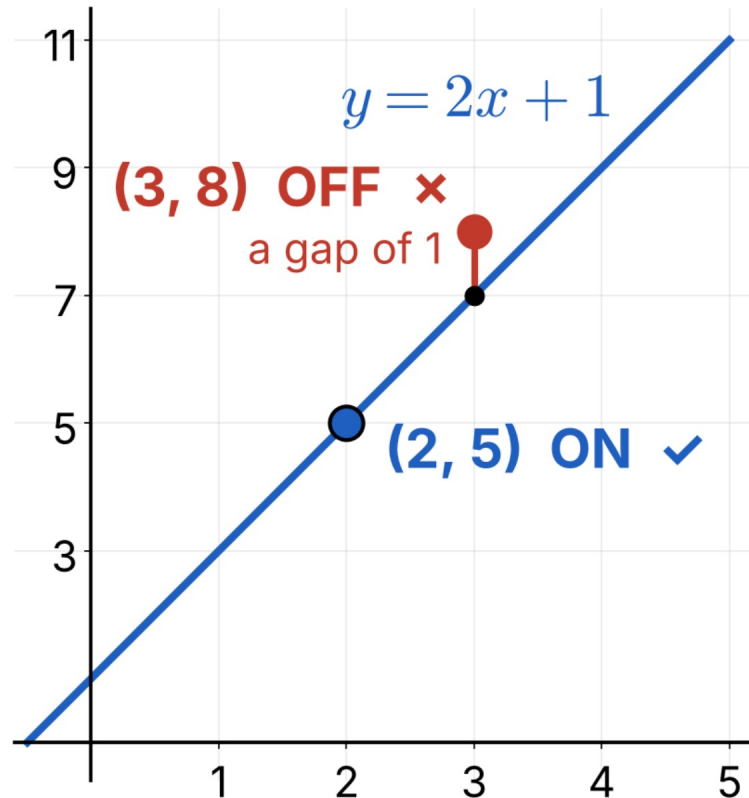
```
import matplotlib.pyplot as plt    # the pen (drawing only)
def g(x):
    return x * x

xs = [-2, -1, 0, 1, 2]           # inputs: how far ACROSS
ys = []                          # outputs: how far UP
for x in xs:
    ys.append(g(x))              # run the machine, keep the answer

plt.scatter(xs, ys)              # one dot per (x, y) pair
plt.plot(xs, ys)                # or: join the dots with lines
```

Two lists, side by side. **xs** is across, **ys** is up. matplotlib only draws; the math is ours.

On the graph, or off it?



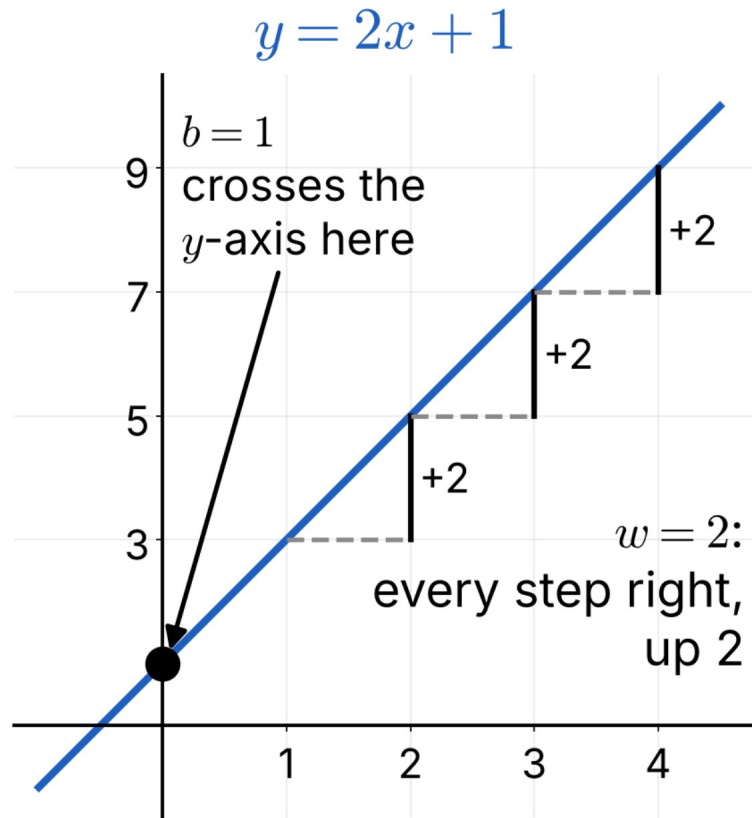
Plug in the **x**. Does the machine give back the **y**?

$$(2, 5) : f(2) = 2 \cdot 2 + 1 = 5 \quad \checkmark$$

$$(3, 8) : f(3) = 2 \cdot 3 + 1 = 7 \neq 8$$

The red gap is a **miss**: how far a dot sits from the graph.

The parts of a line



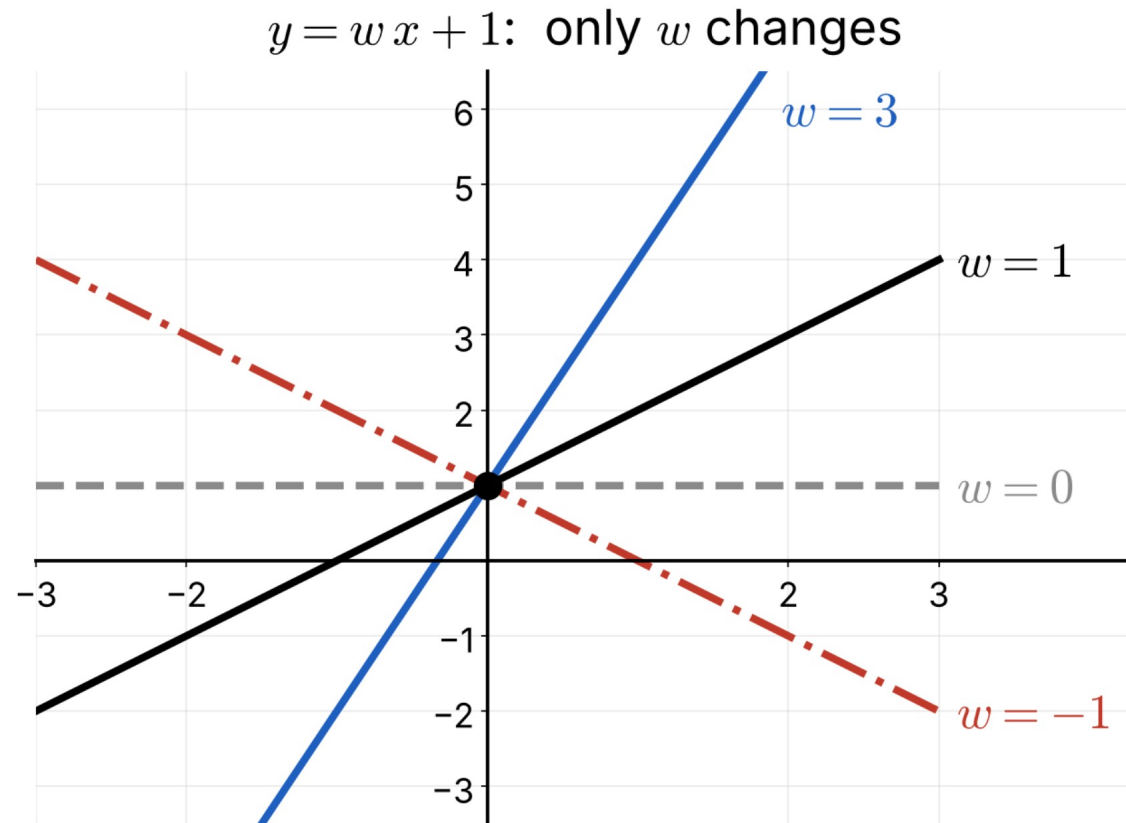
$$y = wx + b$$

b: where the line crosses the y-axis

$$x = 0: \quad y = w \cdot 0 + b = b$$

w: each step right, the line climbs **w**

Turning w tilts the line



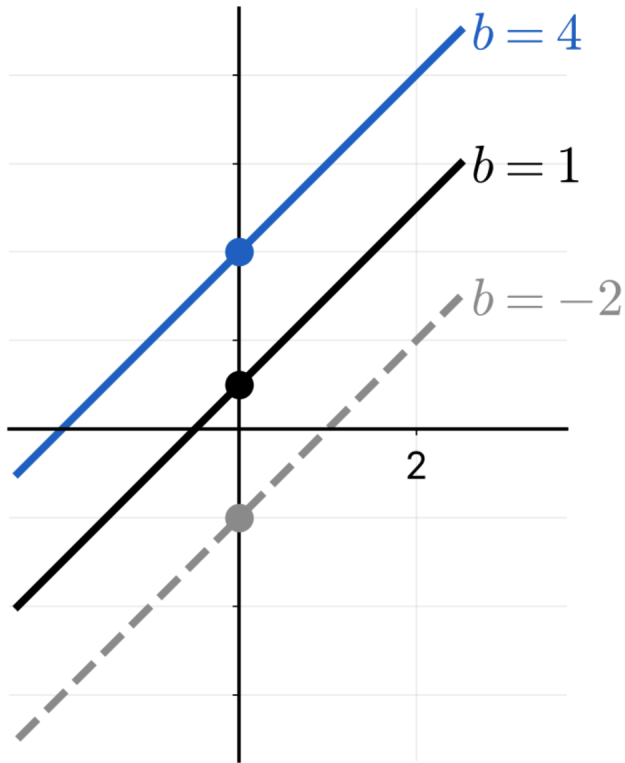
b STAYS AT 1. ONLY w CHANGES.

w	what the line does
3	steep uphill
1	gentle uphill
0	flat
-1	downhill

Farther from 0 = steeper. All cross at $(0, 1)$: that's **b**'s job.

Turning b slides the line

$y = 2x + b$: only b changes



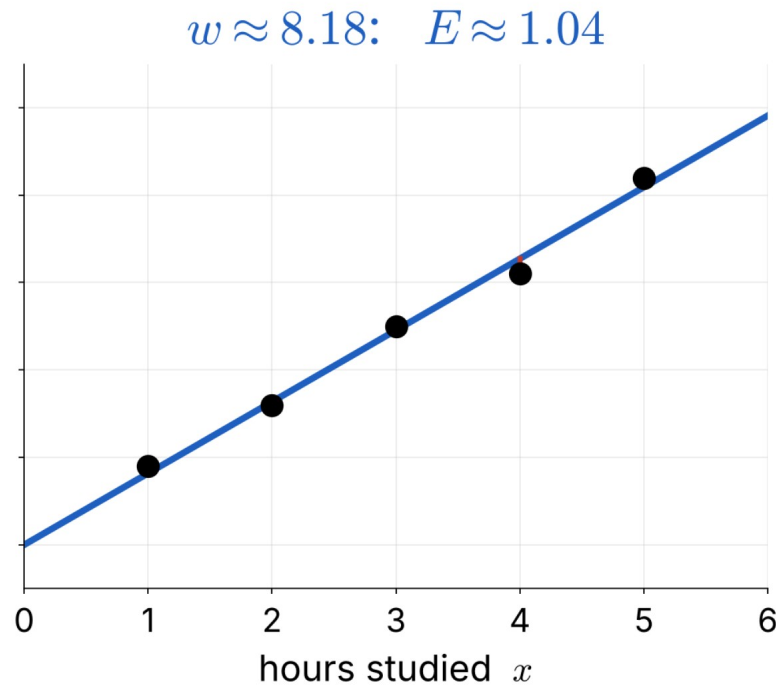
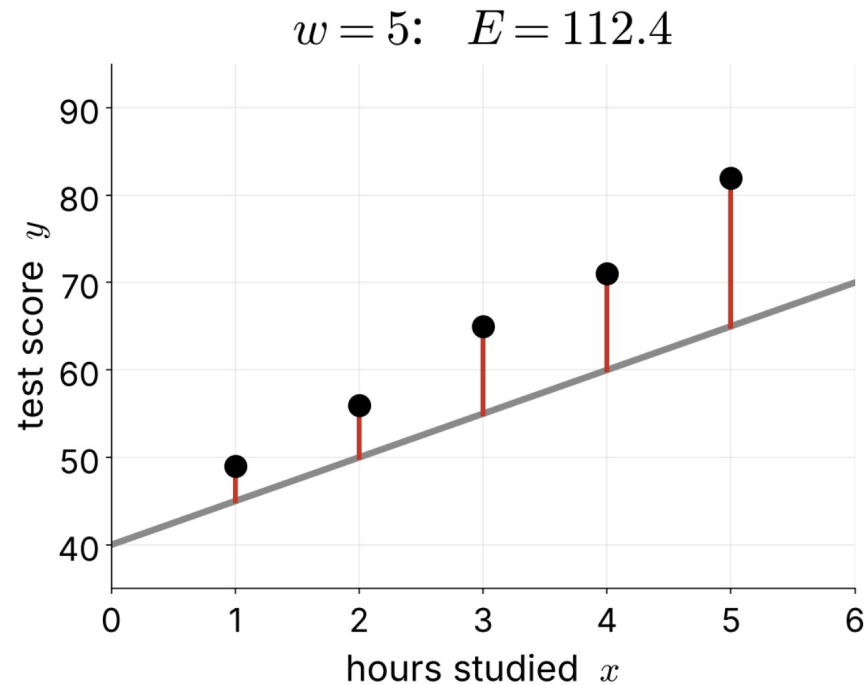
w STAYS AT 2. ONLY b CHANGES.

Same tilt. Different height.

Each line crosses the y-axis at its **b** : 4, 1, -2.

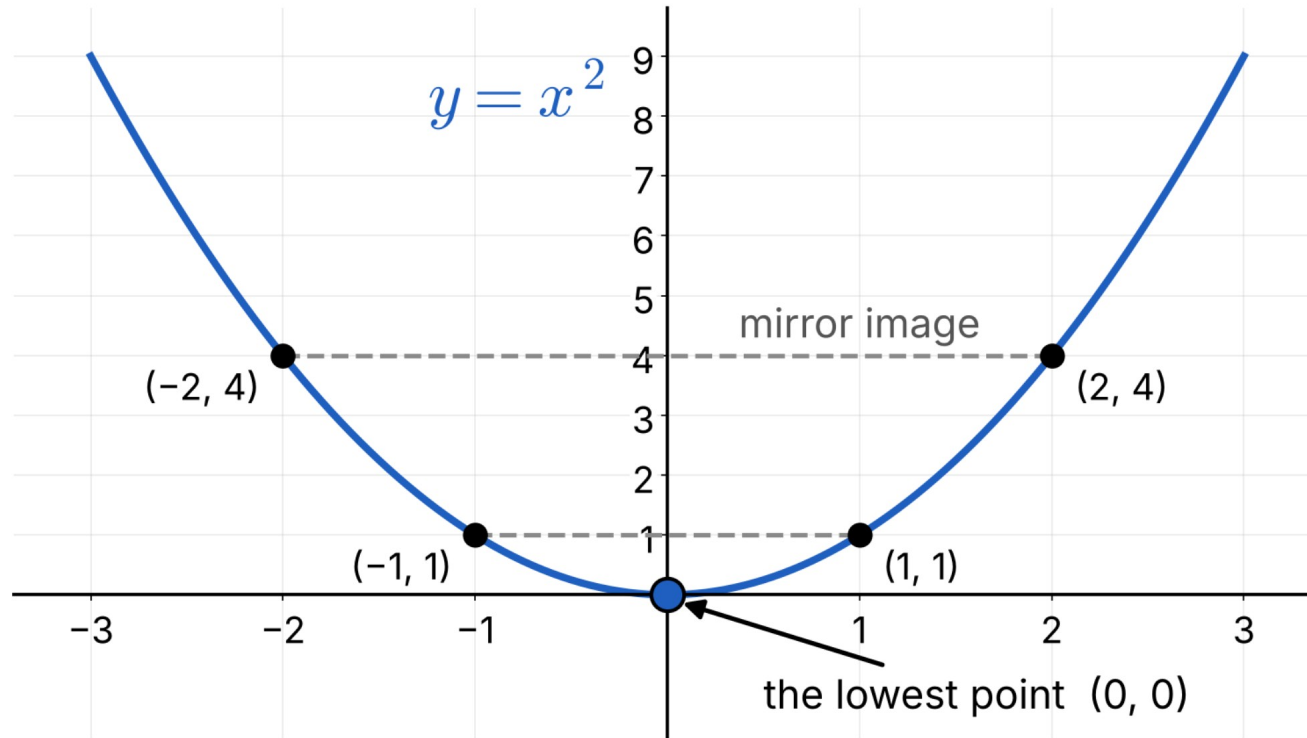
The lines never meet. They're **parallel**.

Our five students, graphed



$\hat{y} = wx + 40$ Red lines are the **misses**. Training turns **w** until they're as small as they can get.

The parabola



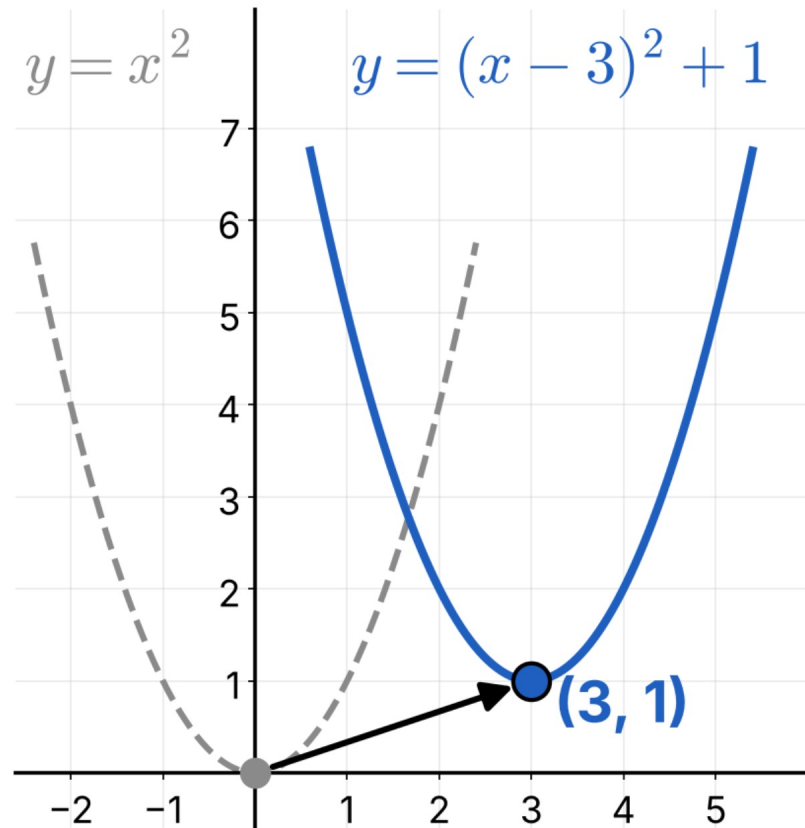
$$y = x^2$$

Mirror image: x and $-x$ land at the same height.

$$(-2)^2 = 2^2 = 4$$

One lowest point at (0, 0).

Moving the parabola



$$y = (x - 3)^2 + 1$$

Inside the parentheses: -3 moves it **right** 3.
(Yes, minus means right. It's a trap!)

Outside: $+1$ moves it **up** 1.

New bottom: **(3, 1)**, because $x = 3$ makes the square 0.

The error valley is a parabola

$$E(w) = \frac{1}{5} \sum_{i=1}^5 [y_i - (w x_i + 40)]^2$$

TIDIES UP TO

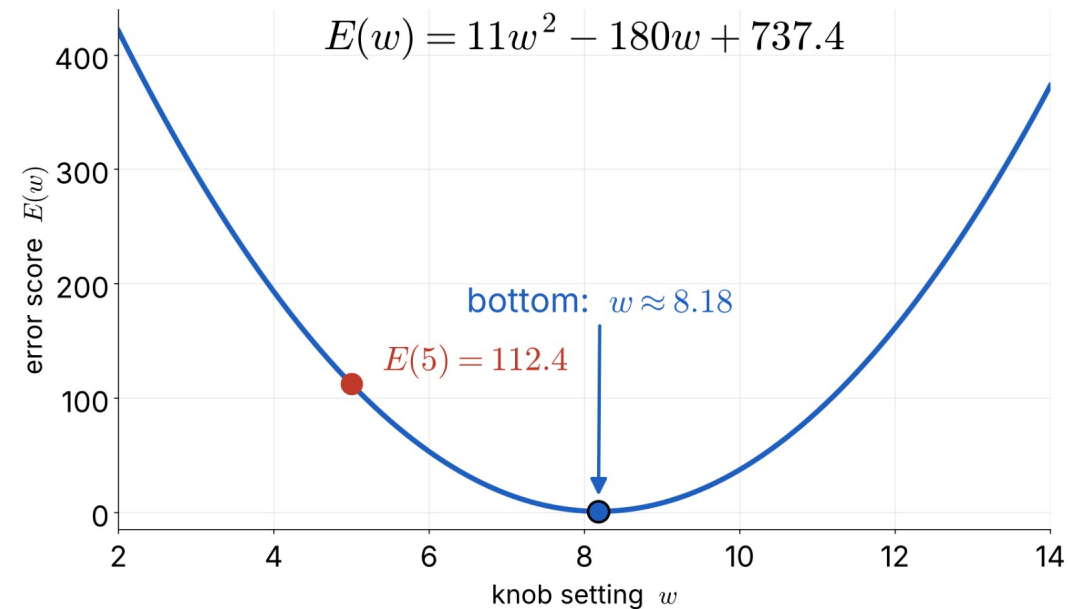
$$E(w) = 11w^2 - 180w + 737.4$$

$$E(5) = 11 \cdot 25 - 180 \cdot 5 + 737.4 = 112.4$$

THE BOTTOM: HALFWAY BETWEEN TWO EQUAL HEIGHTS

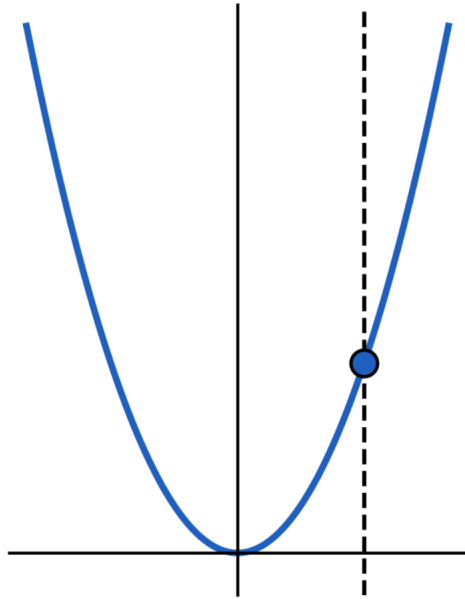
$$E(0) = E\left(\frac{180}{11}\right) = 737.4$$

$$w^* = \frac{0 + 180/11}{2} = \frac{90}{11} \approx 8.18$$

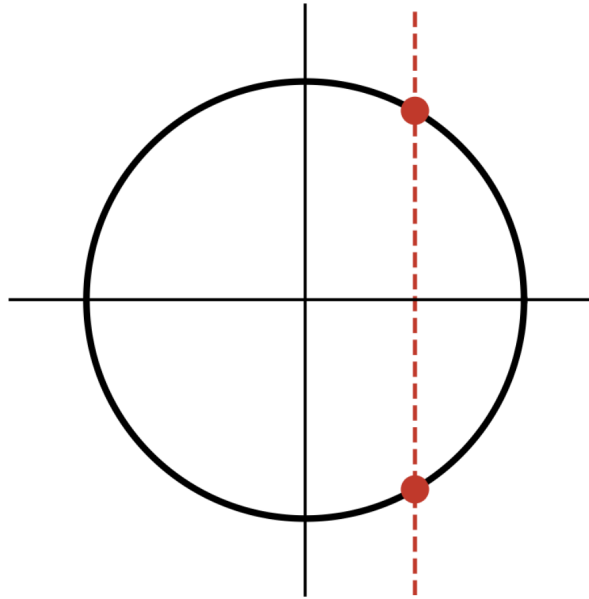


The vertical line test

A function ✓
every vertical line: at most one hit

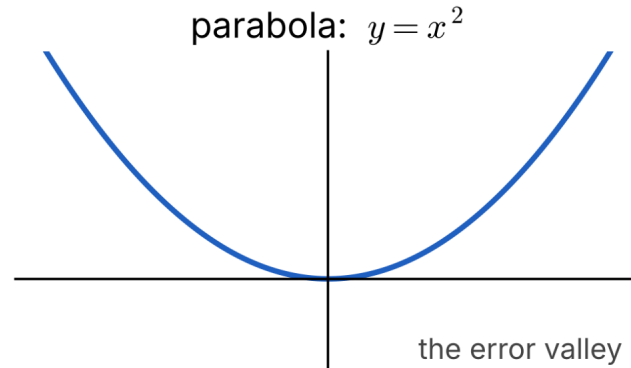
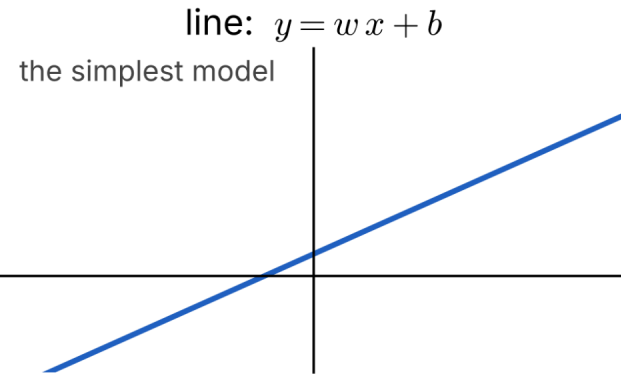


Not a function ×
one input, two outputs



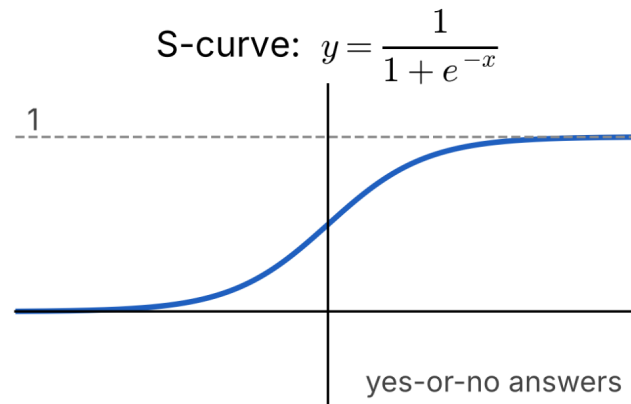
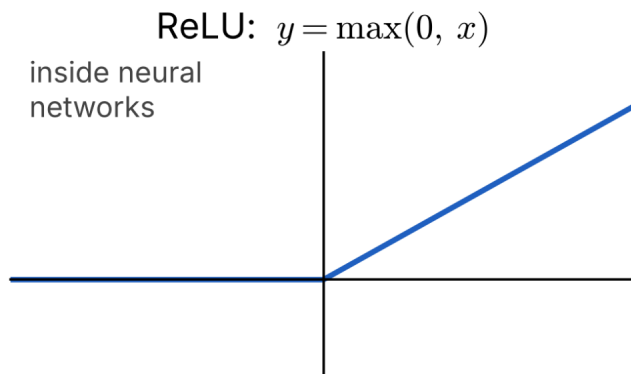
Episode 0.3's rule, as a picture: **one input, exactly one output**. Any vertical line may hit a function's graph at most once. (At most, because some inputs might not be allowed at all.)

Four shapes you'll meet in AI



Line: changes by the same amount every step

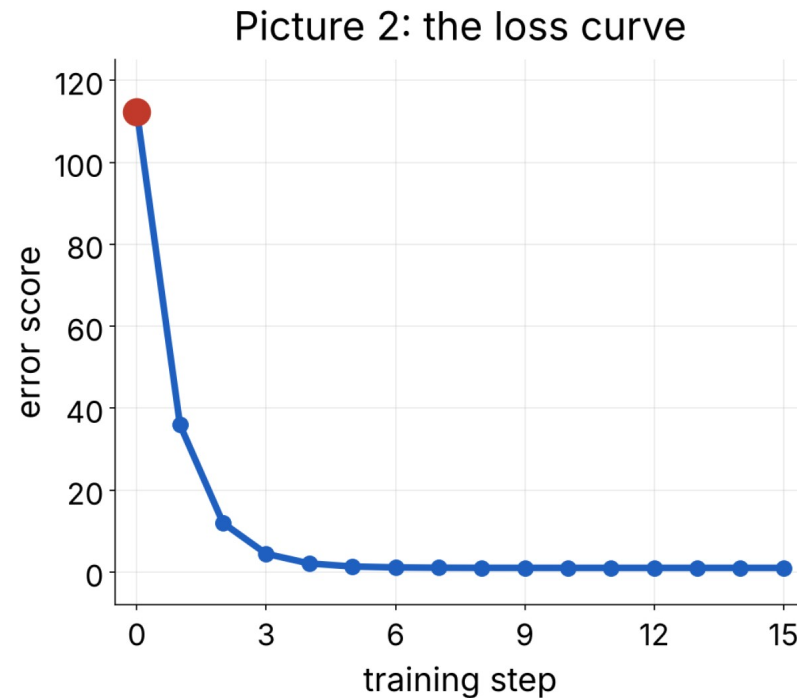
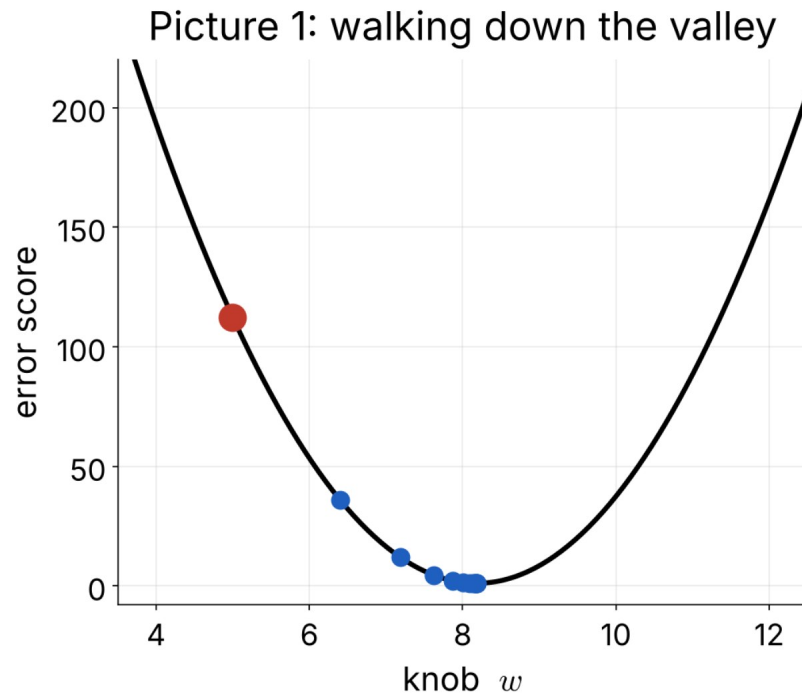
Parabola: an error valley with one bottom



ReLU: flat, then a ramp (Ep 0.3)

S-curve: squashes any number to between 0 and 1

Two pictures of the same training



The **loss curve** (right) is the graph AI engineers watch while a model trains. Going down = the model fits the data better.

Your turn: read the graphs

1 $f(x) = 3x - 1$ $(4, 11)$ Is $(4, 11)$ on the graph?

2 $y = 4x + 5$ Where does this line cross the y-axis?

3 $y = (x - 2)^2 + 5$ Where is the bottom of this U?

Pause the video. Say each one out loud, then work it out.

Answers

1 $f(4) = 3 \cdot 4 - 1 = 11$ ✓ Yes, it's on the graph.

2 $y = 4 \cdot 0 + 5 = 5 \Rightarrow (0, 5)$ It crosses at $b = 5$.

3 $(2, 5)$ $x = 2$ makes the square 0, then +5 lifts it to 5.

THE EQUATION OF THE DAY

$$y = wx + b$$

w tilts the line. b slides it up and down. Training turns the knobs until the line fits the data.

NEXT TIME · EPISODE 0.5

Slope

Rise over run, and the seed of all of calculus.

YOUR CHALLENGE · answer in Episode 0.5

$$y = 2x + 1 : \quad (1, 3) \text{ and } (4, 9)$$

How far up? How far across? Divide up by across. Where have you seen that number before?